Experiment 2 - Statistical Significance

top-k = 4

temp = 0.1; top-p = 0.1

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	P-value > 0.05	N/A			
BioBERT	P-value > 0.05	P-value > 0.05	N/A		
MedEmbed	P-value > 0.05	P-value > 0.05	P-value > 0.05	N/A	
pubmedbert	P-value > 0.05	P-value > 0.05	P-value > 0.05	P-value > 0.05	N/A

temp = 0.05; top-p = 0.01

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0158	N/A			
BioBERT	0.0405	0.7194	N/A		
MedEmbed	0.0528	0.5413	1.0000	N/A	
pubmedbert	0.0591	0.4049	0.8388	1.0000	N/A

temp = 0.005; top-p = 0.001

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0117	N/A			
BioBERT	0.0309	0.7194	N/A		
MedEmbed	0.0405	0.5413	1.0000	N/A	
pubmedbert	0.0450	0.4049	0.8388	1.0000	N/A

top-k = 8

temp = 0.1; top-p = 0.1

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0506	N/A			
BioBERT	0.4185	0.1547	N/A		
MedEmbed	0.1056	0.5839	0.3768	N/A	
pubmedbert	0.0231	1.0000	0.1002	0.4237	N/A

temp = 0.05; top-p = 0.01

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0212	N/A			
BioBERT	0.1853	0.2002	N/A		
MedEmbed	0.0489	0.5708	0.4725	N/A	
pubmedbert	0.0131	1.0000	0.1336	0.4237	N/A

temp = 0.005; top-p = 0.001

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0161	N/A			
BioBERT	0.1486	0.2002	N/A		
MedEmbed	0.0372	0.5708	0.4725	N/A	
pubmedbert	0.0097	1.0000	0.1336	0.4237	N/A

top-k = 12

temp = 0.1; top-p = 0.1

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0077	N/A			
BioBERT	0.1244	0.4825	N/A		
MedEmbed	0.2031	0.0227	0.6527	N/A	
pubmedbert	0.0684	0.2482	0.9075	0.4049	N/A

temp = 0.05; top-p = 0.01

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0029*	N/A			
BioBERT	0.0620	0.5611	N/A		
MedEmbed	0.1056	0.0227	0.5762	N/A	
pubmedbert	0.0336	0.2482	1.0000	0.4049	N/A

temp = 0.005; top-p = 0.001

	Baseline	stella-en	BioBERT	MedEmbed	pubmedbert
Baseline	N/A				
stella-en	0.0021*	N/A			
BioBERT	0.1098	0.3143	N/A		
MedEmbed	0.0824	0.0227	0.9110	N/A	
pubmedbert	0.0256	0.2482	0.8220	0.4049	N/A

Cochran's Q tests and post-hoc pairwise McNemar tests' P-values comparing across different RAG pipelines, with the generation model being *Llama-3.1-8B-Instruct*. P-values smaller than 0.0001 are reported as "< .0001".

Experiment 5 - Statistical Significance

Bare

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.0450	N/A					
Llama-3.2-3B-Instruct	0.0549	0.0005*	N/A				
Llama-3.3-70B-Instruct	0.0016*	0.1058	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	N/A		
Qwen3-8B	P-value < 0.0001*	0.0001*	P-value < 0.0001*	0.0269	0.0689	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.8011	0.0233	N/A

Direct

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.1859	N/A					
Llama-3.2-3B-Instruct	P-value < 0.0001*	P-value < 0.0001*	N/A				
Llama-3.3-70B-Instruct	0.0104	0.1884	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	0.0001*	0.0039	P-value < 0.0001*	0.0890	N/A		
Qwen3-8B	0.0101	0.1360	P-value < 0.0001*	0.8711	0.0689	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0010*	0.8011	0.0233	N/A

Combined

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.0125	N/A					
Llama-3.2-3B-Instruct	P-value < 0.0001*	P-value < 0.0001*	N/A				
Llama-3.3-70B-Instruct	0.0104	0.1884	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	0.0001*	0.0039	P-value < 0.0001*	0.0890	N/A		
Qwen3-8B	0.0101	0.1360	P-value < 0.0001*	0.8711	0.0689	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0010*	0.8011	0.0233	N/A

Easy

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.4610	N/A					
Llama-3.2-3B-Instruct	P-value < 0.0001*	P-value < 0.0001*	N/A				
Llama-3.3-70B-Instruct	0.1949*	0.5611	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	N/A		
Qwen3-8B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.3613	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.6892	0.1237	N/A

Medium

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.1244	N/A					
Llama-3.2-3B-Instruct	P-value < 0.0001*	P-value < 0.0001*	N/A				
Llama-3.3-70B-Instruct	0.0025	0.0778	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0003*	N/A		
Qwen3-8B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0025	0.6265	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0171	0.0010*	N/A

Hard

	Llama-3-8B-Instruct	Llama-3.1-8B-Instruct	Llama-3.2-3B-Instruct	Llama-3.3-70B-Instruct	Qwen2.5-72B-Instruct	Qwen3-8B	Qwen3-32B
Llama-3-8B-Instruct	N/A						
Llama-3.1-8B-Instruct	0.2453	N/A					
Llama-3.2-3B-Instruct	P-value < 0.0001*	P-value < 0.0001*	N/A				
Llama-3.3-70B-Instruct	0.0682	0.4701	P-value < 0.0001*	N/A			
Qwen2.5-72B-Instruct	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	N/A		
Qwen3-8B	0.0070	0.0853	P-value < 0.0001*	0.1763	P-value < 0.0001*	N/A	
Qwen3-32B	P-value < 0.0001*	P-value < 0.0001*	P-value < 0.0001*	0.0010*	0.8744	P-value < 0.0001*	N/A

Cochran's Q tests and post-hoc pairwise McNemar tests' P-values comparing across different model with various size (3/8/32/70/72 billion parameters) and architectures (conventional .vs reasoning types) pipelines, with the embedding model being *gte-Qwen2-1.5B-instruct*. P-values smaller than 0.0001 are reported as "< .0001".